Dev Prakash Jha

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RESEARCH PROFILE

Applied mathematician working at the interface of control theory, dynamical systems, partial differential equations, and scientific machine learning. My research focuses on observability and null controllability of parabolic systems with delay and memory, Carleman estimates, and controllability of degenerate and non-autonomous PDEs. I also investigate approximation, controllability, and robustness of semi-autonomous neural ordinary differential equations. My broader interests concern mathematical analysis of dynamical systems and connections between control-theoretic methods and learning-based models.

RESEARCH INTERESTS

- Control and observability of partial differential equations
- Null controllability of parabolic systems with delay and memory
- Carleman estimates and moving control regions
- Degenerate and non-autonomous parabolic equations
- Scientific machine learning and semi-autonomous neural ODEs
- Approximation, robustness, and controllability of nonlinear dynamical systems
- Fractional evolution equations and operator-theoretic methods
- Numerical analysis and computational methods for evolution equations

EDUCATION

Ph.D. in Mathematics — Indian Institute of Space Science and Technology (IIST), India
Expected completion: January 2027 · CGPA: 8.7/10
Supervisor: Prof. Raju K. George
Research: control theory of PDEs, fractional evolution equations, delay systems, and controllability.

M.Sc. in Mathematics — Indian Institute of Technology Jodhpur (IIT Jodhpur), India
CGPA: 8.15/10 · Third topper.

B.Sc. in Mathematics — Tilka Manjhi Bhagalpur University (TMBU), India
73% · First topper.

SELECTED PUBLICATIONS

Published

1. Dev Prakash Jha and Raju K. George, “Existence and Uniqueness of Mild Solutions and Evolution Operators for a Class of Non-Autonomous Conformable Fractional Semi-Linear Systems and Their Exact Null Controllability.” *International Journal of Dynamics and Control*, 2025. DOI: 10.1007/s40435-025-01844-x.

Accepted for publication

2. Dev Prakash Jha and Raju K. George, “Observability and Null Controllability for Parabolic Systems with Delay and Memory.” *SIAM Journal on Control and Optimization (SICON)*, accepted for publication, 2026.
3. Dev Prakash Jha and Raju K. George, “Memory Obstructs Null Controllability: Two Dichotomies for Linear Systems with Delay and Memory Effects.” *International Journal of Systems Science*, accepted, 2026.

Revision submitted

4. Dev Prakash Jha and Raju K. George, “Geometric Characterization of Memory-Type Null Controllability for Parabolic Equations with Moving Controls.” *Mathematics of Control, Signals, and Systems*, revision submitted, 2026.

PREPRINTS AND CURRENT SUBMISSIONS

1. Dev Prakash Jha and Raju K. George, “Controlled Semiautonomous Neural Ordinary Differential Equations for Approximation and Controllability of Nonlinear Control-Affine Systems.” Submitted to IEEE Transactions on Neural Networks and Learning Systems.
2. Dev Prakash Jha and Raju K. George, “Turnpike-Preserving Universal Approximation for Controlled Semiautonomous Neural ODEs.” Submitted to IEEE Transactions on Automatic Control; under review according to the supplied CV.
3. Dev Prakash Jha, Akambadath K. Nandakumaran, and Raju K. George, “Memory-Type Null Controllability for Non-Autonomous Degenerate Parabolic Equations in Non-Divergence Form.” Submitted to Journal of Differential Equations.
4. Dev Prakash Jha, Akambadath K. Nandakumaran, and Raju K. George, “Memory-Type Null Controllability for Non-Autonomous Degenerate Parabolic Equations in Divergence Form.” Submitted to Journal of Evolution Equations.
5. Dev Prakash Jha and Raju K. George, “Approximate Controllability of Fractional Evolution Equations with Nonlocal Conditions via Operator Theory.” arXiv:2501.17652 (2025); submitted to Fractional Calculus and Applied Analysis, under review according to the supplied CV.
6. Dev Prakash Jha and Raju K. George, “An Operator-Theoretic Framework for the Optimal Control Problem of Nonlinear Caputo Fractional Systems.” arXiv:2504.09611 (2025); submitted to Journal of Optimization Theory and Applications.
7. Dev Prakash Jha and Raju K. George, “Exact Null Controllability of Non-Autonomous Conformable Fractional Semi-Linear Systems with Nonlocal Conditions.” arXiv:2409.16087 (2024).

RESEARCH EXPERIENCE AND PROJECTS

- Controllability and observability for parabolic systems with delay and memory.
- Carleman estimates and controllability with moving control regions.
- Null controllability of degenerate parabolic equations, including time-dependent diffusion and boundary degeneracy.
- Fractional evolution systems with nonlocal conditions and operator-theoretic analysis.
- Semi-autonomous neural ODEs: approximation and controllability of nonlinear control-affine systems; robustness and numerical evaluation.

TECHNICAL SKILLS

Programming and scientific computing: Python, MATLAB, C/C++.
Numerical methods: simulations for dynamical systems and PDEs.
Scientific writing: LaTeX.
Machine learning: teaching-assistant experience in Foundations of Machine Learning and neural ODE research.

TEACHING EXPERIENCE

- Teaching Assistant, Foundations of Machine Learning (MA618), IIST.
- Teaching Assistant, Calculus (MA111), IIST.
- Teaching Assistant, Linear Algebra, Integral Transforms, and Calculus of Variations (MA221C), IIST.

AWARDS AND FELLOWSHIPS

- CSIR–UGC NET with Junior Research Fellowship (JRF), Mathematical Sciences, 2024; All India Rank 153.
- CSIR–UGC NET, Mathematical Sciences, 2021; All India Rank 51.
- GATE Mathematics, 2021; score 405.
- IIT-JAM Mathematics, 2018; All India Rank 350.
- Senior Research Fellowship; Institute Fellowship; Travel Grants.

WORKSHOPS AND SCHOOLS

- Recent Developments in Applied and Computational Mathematics (RDACM), IIST, 2026.
- Recent Advances on Control Theory of PDE Systems (RACP), ICTS, 2024.
- NCM Workshop on Control Theory for Partial Differential Equations, IISER Thiruvananthapuram, 2023.

- 85th Annual Conference of the Indian Mathematical Society, IIT Kharagpur, November 2019.

PROFESSIONAL MEMBERSHIPS

Society for Industrial and Applied Mathematics (SIAM); American Mathematical Society (AMS).

REFERENCES

Prof. Raju K. George (PhD supervisor), Department of Mathematics, IIST, Thiruvananthapuram, India. Email: george@iist.ac.in; rkg.iist@gmail.com.

Prof. Akambadath K. Nandakumaran (research collaborator), Indian Institute of Science (IISc), India. Email: nands@iisc.ac.in.